HepatoTwin

HepatoTwin: A Digital Twin Framework for Personalized, Evidence-Based Dietary Intervention and Health Forecasting

1. Research Question

> **Central question**
>
> Can a patient-specific liver digital twin — built from clinical and biological evidence — support the evidence-grounded evaluation and ranking of existing recipes from a recipe corpus, according to their health relevance, cuisine compatibility, and cooking feasibility for an individual patient?

Clinical focus: metabolic dysfunction-associated steatotic liver disease (MASLD; formerly NAFLD) — diet-modifiable, with strong clinical evidence linking dietary and weight changes to measurable outcomes, giving the project a concrete, gradable success criterion. This is stated as a focus, not baked into the title, because the underlying framework — the twin's state model and its recipe evaluation and ranking engine — is domain-general and could be re-validated against any diet-modifiable condition with a comparable evidence base.

1.1 What Makes This a Digital Twin

A digital twin is a patient-specific computational representation of a system's state that can be updated as new patient data becomes available and used to simulate how that state may change under different interventions — not a one-shot prediction paired with a static visualization (Katsoulakis et al., 2024). HepatoTwin is designed to meet that bar:

> **`Digital Twin ≠ 3D liver`**
>
> The 3D render (Layer 6, §9) is one interface into the twin, not the twin itself. The actual digital twin is the combination of patient state, clinical prediction, biological knowledge, intervention evidence, the scenario engine, and the future-state representation — Layers 2 through 6 together.

- State, not a snapshot: the twin holds a calibrated hepatic risk state plus a nutritional-exposure profile per patient, both of which update on new clinical data or a new recipe plan — not recomputed once and displayed.
- Simulations of future states, not lookups: The scenario engine (§8) is not looking up a static recommendation — it simulates the future state of the twin, given a specific recipe, based on effect sizes derived from the evidence rather than the risk model’s extrapolation.

Scope Note – longitudinal validation not done

The current project does not perform longitudinal validation by comparing the projections to the true repeat measurements outcome data — the clinical data set (§13) used is a cross-sectional one. All projections of the future states are marked as “simulated projections, not a clinical prediction” and all scenarios are based on the clinical evidence (§6).

2. System Architecture — Seven Layers

The digital twin is the whole stack, not any single layer — Layer 4, the recipe evaluation and ranking engine, is the primary research contribution and is referred to internally as NutriTwin.

Layer	Contents	Role
1. Patient State	Demographics, lifestyle, clinical biomarkers, current diet (as a recipe list)	Raw input
2. Liver Digital Twin — Current State	Calibrated MASLD/steatosis risk, fibrosis risk (FIB-4, §17.2), biological state representation, causal knowledge graph	The twin's core (§5)
3. Intervention Evidence Engine	Clinical trials/meta-analyses, causal-graph gating	Evidence-graded health signal (§6)
4. Recipe Evaluation and Ranking Engine (NutriTwin)	Health, cuisine, and cooking-feasibility objectives over RecipeDB	Core research contribution (§7)
5. Scenario Engine	"What if the patient adopts Recipe A / B / C?" → evidence-graded projected direction	Makes the twin interactive (§8)
6. Digital Twin Visualization	3D liver (simplified, S0–S3) + timeline slider (today → 3m → 6m → 1y)	The twin's interface (§9)
7. Explainable Dashboard	Why this recipe, evidence sources, nutritional fit, confidence	User-facing layer (§10)

3. Need for the following Databases

Layers 3 and 4 are where the lab's own assets are load-bearing:

Lab asset	Role in this project
RecipeDB	Source corpus for candidate recipes; supplies ingredient/process/utensil structure used for the cooking-feasibility objective.
FlavorDB	Compound-level ingredient flavor profiles; explored during the literature review as a possible flavor-similarity signal, but not used as a core scoring component, since the flavor of a fully cooked, multi-ingredient dish cannot be reliably predicted from individual compound profiles (see §4).
Computational gastronomy methodology	The recipe evaluation methodology itself — multi-criteria scoring and ranking of structured recipe data, linked to an evidence table — extends the food-pairing / recipe-recommendation line of work already established in the lab's own literature (§4).

The clinical model (Layer 2) is deliberately scoped as an input to the twin, not the object of study — the thing being invented is the ranking engine, not the risk predictor.

4. Related Work — Where the Gap Sits

Three related strands of research point to the open problem addressed in this work:

- **Computational gastronomy and recipe databases:** Resources such as RecipeDB provide structured information about recipes, including ingredients, nutritional profiles, cuisines, cooking processes, dietary styles, and associations with flavor and health-related resources. However, these systems are primarily designed for exploring and querying culinary data rather than evaluating existing recipes for the specific needs of an individual patient.
- **Food, nutrition, and health knowledge resources:** Clinical trials, meta-analyses and practice guidelines connect dietary interventions with measured liver outcomes. These resources provide valuable evidence that can support personalized dietary decisions. However, they do not by themselves provide a framework for evaluating complete recipes according to a patient's current health state.
- **Patient-specific metabolic and liver digital twins:** Existing work on digital twins and personalized nutrition focuses on modeling patient state, biomarkers, disease progression, and personalized interventions. However, existing recipes are generally not treated as a structured candidate space that can be systematically evaluated according to multiple criteria relevant to an individual patient.

These three areas provide complementary pieces of the problem: recipe databases provide structured descriptions of food and recipes, health knowledge resources provide evidence about food–health relationships, and patient-specific digital twins model the health state of an individual. However, these components are typically studied separately.

Recent work shows the area is active but early. In an open-label randomised trial in 319 people with type 2 diabetes, digital-twin-guided nutrition, which selected foods by their predicted post-meal glucose response, improved liver-fat scores and MRI-PDFF over one year (Joshi et al., 2023). The Horizon Europe project ARTEMIS (grant 101136299; registered as the observational study NCT07430501) is building virtual twins of MASLD from retrospective data on about 7,700 patients across European centres, including prediction of cardiovascular events. Neither ranks existing recipes against evidence-graded, patient-specific objectives.

The Research Gap

Based on our literature review, we identify a gap at the intersection of **patient-specific health modeling, evidence-based nutrition, and computational gastronomy**. While structured recipe databases allow exploration of existing recipes, and digital twin approaches support patient-specific health modeling, there is limited work connecting these components to systematically evaluate and rank existing recipes for a particular patient's health state.

In particular, we did not identify a framework that combines:
health relevance, derived from the patient's current state and evidence-based dietary objectives;
cuisine compatibility, to account for the patient's dietary and culinary context; and
practical cooking feasibility, based on recipe ingredients and cooking requirements;
within a transparent multi-criteria framework for patient-specific recipe evaluation.

How HepatoTwin Addresses the Gap

Rather than generating new recipes or performing ingredient substitutions, **HepatoTwin evaluates existing candidate recipes** from a structured recipe corpus. The patient's current liver and metabolic state is first represented through the digital twin, which is then used to derive evidence-based dietary objectives.

Each candidate recipe is subsequently evaluated according to its **health relevance, cuisine compatibility, and practical cooking feasibility**. These criteria are combined using a transparent multi-criteria scoring framework to produce a ranked list of the most suitable recipe options for the patient.

Thus, the central contribution of HepatoTwin is not recipe generation or flavor-preserving ingredient substitution. Instead, it provides a framework for connecting a **patient-specific liver digital twin with evidence-based dietary knowledge and structured recipe data to support transparent evaluation and ranking of existing recipes**.

5. Layer 2 — Liver Digital Twin Core (Current State Model)

5.1 Steatosis / MASLD risk model

- XGBoost main model, compared to linear regression and random forest baselines
- Trained and evaluated on one cohort only (§13) to prevent cross-population validation bias
- Postprocessed for proper calibration using Platt scaling/isotonic regression; reported together with Accuracy, F1, ROC-AUC as well as with the calibration plot and Brier score

5.2 Explainability (SHAP)

SHAP determines which input features contributed most to a particular prediction of the current state, thus providing an interpretation at the level of features of the “current state” of the twin prior to intervention.

5.3 Causal knowledge graph

A directed graph containing only a handful of hand-crafted relationships (~10-15), representing the well-known causal directionality of MASLD biology — for example, dietary/insulin resistance -> fat deposition in the liver -> consequent ALT/AST increase. This will limit the types of mutations that can be scored in Layer 3 to those that are plausibly causal within the scope of the problem definition, and ensure that the health objective does not try to act on a downstream consequence.

6. Layer 3 — Intervention Evidence Engine

Every intervention effect used by the ranking engine is pulled from a named clinical trial or meta-analysis, with an explicit evidence-strength label. The current table has 23 rows, each with an evidence ID cited by the rules; a selection is shown:

Intervention	Outcome measured	Reported effect	Source	Evidence strength

Weight loss ≥5% of body weight	Hepatic steatosis (biopsy)	Reduction in steatosis observed at this threshold; dose-response with further loss	Vilar-Gomez et al., Gastroenterology, 2015	Strong (n=293, biopsy-paired)
Weight loss 7–10%	Liver inflammation / NASH activity	Improvement in NAFLD Activity Score at this threshold	Vilar-Gomez et al., Gastroenterology, 2015	Strong (n=293, biopsy-paired)
Weight loss ≥10%	NASH resolution / fibrosis	90% NASH resolution and dose-dependent fibrosis regression at this threshold	Vilar-Gomez et al., Gastroenterology, 2015	Strong (n=293, biopsy-paired)
Mediterranean diet vs. control	AST, GGT	Small but statistically significant reduction	Sangouni et al., Br J Nutr, meta-analysis of 10 RCTs, 2022	Moderate (meta-analysis, n=705)
Mediterranean diet vs. control	ALT	No statistically significant effect; CI crosses zero	Sangouni et al., Br J Nutr, meta-analysis of 10 RCTs, 2022	Weak / inconclusive
Mediterranean diet vs. low-fat diet	Liver enzymes, liver fat	No significant difference between the two diets	Xiong et al., Food Funct, 2024 (PubMed 39076035)	Moderate (meta-analysis)
Structured exercise (independent of weight loss)	Liver fat content	Meaningful reduction reported vs. control	Stine et al., Am J Gastroenterol, 2023; 14 RCTs, n=551	Moderate (meta-analysis)

Effect directions are cross-checked against AASLD practice guidance and the EASL-EASD-EASO MASLD guidelines, the current clinical reference standard.

Evidence strength system

- STRONG — biopsy-paired or large-RCT evidence, dose-response established
- MODERATE — meta-analysis or single well-powered RCT
- WEAK / INCONCLUSIVE — evidence exists but the effect is not statistically significant, or CIs cross zero
- INSUFFICIENT — no adequate evidence for this specific recipe; the ranking engine excludes or flags it rather than guessing
- GUIDELINE RECOMMENDATION — advice from a current clinical practice guideline (EASL-EASD-EASO 2024), used for its direction only and never converted into an effect size
- OBSERVATIONAL ASSOCIATION — an association from observational studies (for example, coffee and lower odds of fibrosis); never used to set a priority or score a recipe, and reported as an association rather than an effect

6.1 Evidence Granularity and Intervention Mapping

The clinical data is generally documented on the level of threshold for weight loss or diets, but not at the level of an individual recipe. The framework does not assume any extrapolation of a dietary pattern effect to a single recipe, and instead determines which of the intervention categories is associated with a given recipe's nutritional profile, treating a recipe that cannot be mapped to any evidenced category as INSUFFICIENT. The most similar literature is related to identifying food ingredient substitutions using a knowledge graph of food (PMC 2021) and a survey on food ingredient substitutions (arXiv 2025).

7. Layer 4 — Recipe Evaluation and Ranking Engine (NutriTwin, Core Research Contribution)

Based on the patient's condition at Layer 2 and the nutritional objectives derived from Layer 3, evaluate and rank candidate recipes from RecipeDB that would enhance the health relevance predictions while remaining culturally and practically appropriate for the patient

7.1 Evaluation Criteria

Criterion	Information Source	Goal
Health relevance	Layer 3 evidence-graded effect sizes from clinical trials, meta-analyses and guidelines	Higher is better
Cuisine compatibility	Ingredient co-occurrence frequency within the recipe's cuisine cluster in RecipeDB	Prefer recipes compatible with the patient's dietary/cultural context
Cooking feasibility	Predicted prep-time and skill shift, from RecipeDB process/utensil metadata	Prefer recipes within practical constraints

7.2 Recipe Scoring and Ranking

Scoring criteria: Each candidate recipe is evaluated using a set of evidence-based criteria. A health relevance score is calculated from the nutritional characteristics of the recipe and its relationship to evidence-supported dietary interventions for MASLD. Cuisine compatibility measures how well the recipe matches the patient's dietary or cultural context. Cooking feasibility considers practical factors such as preparation time, cooking process, ingredients, and required utensils.

Implementation: These criteria are normalized and combined using a transparent multi-criteria scoring framework with weights explicitly defined and reported (health 0.6, cuisine compatibility 0.2, cooking feasibility 0.2, acceptability 0.1, renormalised). Acceptability is the RecipeDB rating shrunk towards the corpus mean by the number of ratings, so recipes many people have rated well are preferred. A health gate is applied by default: recipes that meet every evidence-based nutritional reference are ranked ahead of those that do not, so a high cuisine or feasibility score cannot compensate for a weak health profile. Among recipes that meet every reference, half of the health score reflects how far a recipe goes beyond the references on the patient’s higher-priority criteria, so the patient’s priority levels continue to shape the ranking. Within each group, recipes are ordered by their overall weighted score, and a sensitivity analysis

examines how changes in the criterion weights affect the ranking. The system returns the highest-ranked recipes together with the individual criterion scores so that the recommendation remains interpretable.

7.3 Explanation

Each recommended recipe is validated by working back through the knowledge graph to the evidence-table rows and strength of evidence that support it, and the cuisine/process context that kept it appropriate for the patient.

8. Layer 5 — Scenario Engine

The patient (or physician) can compare "if I choose Recipe A" versus "if I choose Recipe B," both based on Layer-3 evidence, not querying the Layer-2 engine.

Scenario	Reported as
Current state	`Calibrated model probability from Layer 2, e.g. "68% predicted probability of grade ≥S1, 95% bootstrap interval (§17.4)"`
Scenario A — max health	"Strong evidence of improvement; larger shift from the patient's usual recipes" — evidence: STRONG
Scenario B — closer to usual recipes	"Moderate evidence of improvement; closer to the patient's usual recipes" — evidence: MODERATE
Untabulated intervention	"Insufficient evidence in current table to simulate this scenario"

This remains an evidence-graded simulated projection, not a clinical forecast - no scenario is ever shown as a bare, model-invented percentage.

9. Layer 6 — Digital Twin Visualization

Purposefully kept simple: the freely available pre-made 3D liver mesh (without creating a new one) where the colour of the material depends on the steatosis state and not on time required for modeling the anatomy.

- Colour-coding for the state of liver: Normal (S0) – MILD (S1) – MODERATE (S2) – SEVERE (S3)
- Timeline slider: TODAY – 3 MONTHS – 6 MONTHS – 1 YEAR demonstrating the Layer-5 evidence-based direction under a chosen scenario
- Clear labeling on screen "simulated projection, not clinical forecast" for all views of future states

This visualization is not the digital twin but just one of its interfaces (§1.1). It could be made even more simple in case of tight schedule without affecting the existence of the digital twin.

10. Layer 7 — Explainable Dashboard

- View the patient's current dietary choices alongside recommended existing recipes
- Compare top-ranked recipes side by side, each with health/cuisine/feasibility scores and evidence-strength labels
- Explore the twin's visualization and timeline for a selected scenario
- Read the knowledge-graph-traced explanation for every recommended recipe

11. Integrated Pipeline

- Step 1: Collection & Pre-processing of Patient Data (Layer 1)
- Step 2: Steatosis/MASLD risk prediction, calibration & SHAP (Layer 2)
- Step 3: Evidence-based dietary objectives (Layer 3)
- Step 4: Recipe evaluation and ranking (Layer 4)
- Step 5: Scenario comparison (Layer 5)
- Step 6: Visualization (Layer 6)
- Step 7: Explainable dashboard (Layer 7)

12. Evaluation Plan

12.1 Digital twin state model (Layer 2)

- Held-out test split from a single cohort; calibration curve, Brier score, Accuracy/F1/ROC-AUC
- Benchmark against established clinical scores: for CAP ≥ 248 dB/m the calibrated model reached an ROC-AUC of 0.860, compared with 0.842 for the Fatty Liver Index (Bedogni et al., 2006) and 0.813 for the Hepatic Steatosis Index (Lee et al., 2010) on the same test participants (Appendix A).

12.2 Recipe evaluation and ranking engine (Layer 4)

- The ranking engine was evaluated on 300 held-out test patients by the share of top-10 recipes that meet all three criteria: every evidence-based nutritional reference for the patient (energy, saturated fat, sugars and fibre), the patient’s cuisine preference, and practical limits of 60 minutes, 15 ingredients and 12 steps. The default health-gated ranking reached 0.83 (0.80 without acceptability), and was the only method whose recommended recipes always met the nutritional references; weighted scoring alone reached 0.73, health-score-only and nutrition-only rankings 0.02, and random candidate selection 0.003. Because NHANES does not record cuisine preferences, these were simulated (Appendix B).
- A sensitivity analysis over 36 weight combinations found a median top-10 overlap of 0.78 with the default ranking; overlap fell most when the cuisine weight was reduced to 0.1.
- Finally, where feasible, a small qualitative expert or pilot evaluation may assess whether recommended recipes appear reasonable and appropriate for their intended cuisine and practical context

12.3 Twin trajectories

Not evaluated against real longitudinal outcomes in this project — see the scope note in §1.1. Twin trajectories are reported strictly as evidence-graded simulated projections.

12.4 Evaluation of the extensions (§17)

- Fibrosis triage: discrimination of FIB-4 (and any fibrosis model) for measured liver stiffness ≥ 8 kPa on the held-out NHANES participants, and agreement of the referral category with the guideline pathway.
- Uncertainty: empirical coverage of the conformal CAP intervals (target 90%) on the test set.
- Personalisation: how far recommendations differ between patients with different habitual diets and priorities (top-10 overlap), before and after adding 24-hour recall data.
- Recommendation diversity and meal plans: intra-list ingredient diversity and the share of plans meeting every weekly constraint.
- Plain-language summaries: automatic faithfulness rate (every number and citation traceable to the twin output) and readability, plus review of a sample by the project team.
- Fairness: Layer 2 metrics and Layer 4 criteria satisfaction by sex, race/ethnicity and age group, with 95% bootstrap intervals.
- NHANES III: external discrimination and calibration against graded ultrasound, and association of the twin's risk and FIB-4 with all-cause and liver-related mortality.

13. Dataset Strategy

Dataset	Used for	Notes
NHANES 2017–March 2020 + August 2021–August 2023 (primary)	Clinical training/validation cohort for Layer 2	Liver Ultrasound Transient Elastography (LUX) component provides CAP-based steatosis grading, in the same public release as biomarkers (ALT, AST, HbA1c) and 24-hour dietary recall, linked by participant ID. Public-use files, no application — downloadable directly from the CDC NHANES data portal.
NASH-CRN (optional, parallel)	Secondary validation cohort if time allows	Biopsy-confirmed clinical cohort; access typically a few weeks, applied for in parallel, not blocking
RecipeDB	Recipe corpus, candidate recipes, cuisine/process metadata	Backbone of Layer 4
FlavorDB	Flavor-compound profiles; explored but not used as a core scoring component (see §4, §7)	—
USDA FoodData Central / IFCT	Nutrient look-up for recipe-to-nutrient conversion	Reference table, not patient data

NHANES III (1988–1994) hepatic steatosis ultrasound + Linked Mortality Files (proposed, §17.9)	External validation against graded ultrasound; long-term outcome validation	Public-use; ultrasound re-read released 2011 (adults 20–74); mortality follow-up to 31 December 2019
NHANES 24-hour dietary recall (DR1TOT, DR1IFF) and USDA FPED 2017–March 2020 (§17.1)	Patient's habitual diet; HEI-2020 diet-quality score	Downloaded; FPED for 2021–2023 derived from DR1IFF with the 2017–2020 food-level FPED and validated against USDA totals
International GI tables 2021; FoodProX (§17.10–17.11)	Glycaemic load and ultra-processing criteria	Published reference tables and open model; mapped to recipe ingredients

UK Biobank was considered but not pursued — its longitudinal repeat-imaging subset was attractive, but its access-approval timeline is measured in months and doesn't fit this project's window. MIMIC-IV remains excluded — an ICU population is not a matched cohort for this task. Continuous NHANES cycles from 1999 to 2016 remain excluded because they have no liver imaging. NHANES III (1988–1994) is an exception: archived gallbladder ultrasound was re-read to grade hepatic steatosis, and its participants are linked to mortality through 2019, so it is now proposed as an external and long-term validation cohort (§17.9).

14. Expected Contributions & Publication Targets

- A digital twin for the patient liver, where the state model is built by updating with new information from the patient, along with a simulation model for evidence-based intervention, tested using a cross-sectional analysis on the NHANES dataset; longitudinal validation left for future work
- New scoring approach for the task of evidence-based recipe evaluation and selection (health+cuisine+feasibility), linked to an evidence table and a causal knowledge graph, which drives the intervention engine of the twin
- An evidence-linked recipe recommendation component that explains why each recipe is suitable, including its nutritional characteristics and supporting intervention evidence, along with a small-scale human evaluation experiment inspired by the Turing-Test-for-Chef framework.
- A fibrosis-aware triage step and MASLD subtype labelling, so that diet recommendations follow the guideline care pathway rather than replacing it (§17.2–17.3)
- Open source, modular architecture of the twin, demonstrated and evaluated in the MASLD use case

15. Risks & Mitigations

Risk	**Mitigation**
NHANES is cross-sectional — no repeat-imaging validation	Named explicitly as a scope limitation (§1.1, §12.3); twin trajectories framed as simulated projections throughout, never as validated forecasts

Evidence cannot always be mapped directly to an individual recipe	Use explicit evidence-strength categories and mark unsupported mappings as INSUFFICIENT rather than guessing
Pilot panel hard to recruit even at small scale	5–10 people (friends/family/department contacts who cook the relevant cuisine) is a realistic bar within the timeline; framed honestly as a pilot, not a scaled study
Two-layer integration slips	Thin end-to-end MVP built jointly in the opening weeks surfaces integration issues immediately, not near the deadline
Compressed timeline overall	Weekly Friday grading meetings function as built-in checkpoints; a pre-defined cut list (3D fidelity first, then panel size, then evidence-table breadth) identifies what to trim if behind schedule, without touching the ranking engine itself
Language-model summaries state facts not in the twin output	Generate only from the structured output; automatically check every number and evidence ID against it; fall back to a template summary when the check fails
NHANES III differs from current NHANES (era, ultrasound reference, assays)	Report it as external validation of transportability, map variables explicitly, and recalibrate if needed rather than retraining

16. Tech Stack

Module	**Technology**
Frontend	React + Tailwind CSS
Backend	FastAPI (Python)
Clinical ML	XGBoost, scikit-learn (isotonic calibration on out-of-fold predictions), SHAP values (XGBoost TreeSHAP)
Recipe Evaluation & Ranking	Python, pandas, NumPy/scikit-learn; health-gated, transparent weighted multi-criteria scoring with sensitivity analysis on criterion weights
Knowledge Graph	NetworkX (small, hand-curated graph — no need for a full graph database at this scale)
Twin Visualization	Three.js / React Three Fiber, pre-built free liver mesh, state-bound colour + timeline slider
Database	PostgreSQL or SQLite (given the compressed scope, SQLite may be sufficient)

17. Extensions Identified During Development

Building and evaluating the first end-to-end version of the twin (October 2026) showed where the design could be strengthened. Extensions 17.1–17.8 and 17.12 use data the project already holds (17.1 also uses USDA FPED, now downloaded); 17.9–17.11 and 17.13 need further public datasets. Each states the layer it extends; their evaluation is listed in §12.4.

#	Extension	Layer	Data	Why it matters
17.1	Habitual diet from 24-hour recall	1, 3, 5	NHANES DR1TOT, DR1IFF and USDA FPED (held)	Makes priorities personal; provides the "usual diet" for Scenario B
17.2	Fibrosis risk and referral pathway	2	Age, AST, ALT, platelets; measured stiffness (held)	Follows the guideline care pathway; flags patients who need a clinician
17.3	MASLD subtype and safety guardrails	2, 3	Alcohol, medication, age (held)	Avoids diet-only advice where it is inappropriate
17.4	Patient-level explanations and uncertainty	2, 7	Existing model	"Why is my risk high?" with an honest range
17.5	Re-assessment timeline	1, 2, 6	Measurements entered over time (no new dataset)	Implements the "updating twin" claim
17.6	Diversity and weekly meal plans	4	RecipeDB (held)	Varied, usable weekly plans in the patient's own cuisine
17.7	Grounded plain-language summaries	7	Twin output	Patient-friendly, traceable explanations
17.8	Fairness audit	Evaluation	NHANES (held)	Checks performance across groups
17.9	External and long-term validation	2	NHANES III ultrasound; Linked Mortality Files	First outcome data for the twin; external cohort
17.10	Ultra-processing criterion	4	FoodProX; FNDDS	Implements the guideline advice on ultra-processed foods
17.11	Glycaemic load criterion	4	International GI tables 2021	Supports the glucose-related priority
17.12	Recipe acceptability	4	RecipeDB ratings (held)	Recipes people like
17.13	Additional evidence rows	3	Published meta-analyses	Coffee, fasting and null findings

17.1 Habitual diet from 24-hour dietary recall

Data. NHANES records each participant's intake on the day before the examination (DR1TOT nutrient totals and DR1IFF individual foods with FNDDS codes), in both FibroScan cycles. The USDA Food Patterns Equivalents Database (FPED) gives added sugars and the food groups for a Healthy Eating Index-2020 score (Shams-White et al., 2023). USDA has released FPED only for 2017–March 2020, so for 2021–2023 the totals are derived with USDA's method from the individual foods and the 2017–2020

food-level FPED; applied to 2017–2020, this reproduced USDA's totals (r = 1.00 for added sugars and HEI-2020).

Use. A priority is raised only when the patient's own intake exceeds the reference value used for recipes (for example, saturated fat above 10% of energy) and lowered when it is already met. This addresses the main limitation found in the ranking evaluation: most patients received the same priorities. With the recall, 300 test patients had 35 distinct priority profiles instead of 11 and 29 distinct top-10 lists instead of 16, and patients with a higher saturated-fat or sugar priority received recipes lower in that nutrient. The same data provide the patient's usual diet for Scenario B (§8) and the "current dietary choices" view of the dashboard (§10).

Caveat. A single 24-hour recall is noisy for an individual; it is used as context for prioritisation, never as a diagnosis.

17.2 Fibrosis risk and referral pathway

Method. The FIB-4 index (Sterling et al., 2006) is computed from age, AST, ALT and platelet count, all in the cohort, and categorised using the EASL-EASD-EASO 2024 pathway (below 1.3 low risk, with 2.0 used above age 65; 1.3–2.67 further testing; above 2.67 high risk).

Validation. NHANES measured liver stiffness in the same FibroScan examination (LUXSMED), so FIB-4 and an optional fibrosis classifier can be validated against stiffness of at least 8 kPa, the guideline's transient-elastography threshold for ruling out advanced fibrosis (12 kPa rules it in), as was done for steatosis.

Use. The twin state gains a second dimension, fat and scarring. Patients in the high-risk category see a recommendation to seek clinical assessment before any recipes.

17.3 MASLD subtype and safety guardrails

Subtype. Participants are labelled using the 2023 multisociety nomenclature (Rinella et al., 2023): steatotic liver disease with at least one cardiometabolic criterion is MASLD, and MASLD with weekly alcohol intake of 140–350 g (women) or 210–420 g (men), on average 20–50 g and 30–60 g a day, is MetALD. NHANES alcohol questionnaire items are already in the dataset.

Guardrails. Participants under 18, those with heavy alcohol intake, those taking glucose-lowering medication and those who are pregnant receive an explicit "discuss with a clinician" notice, because adult dietary guidance or unsupervised diet change may not be appropriate.

17.4 Patient-level explanations and uncertainty

Explanations. Per-patient SHAP values identify the measurements that raise or lower each patient's predicted risk, complementing the global SHAP summary.

Uncertainty. Split conformal prediction gives each predicted CAP a distribution-free interval with a target coverage of 90%, checked on the test set; bootstrap intervals accompany the calibrated probabilities.

17.5 Re-assessment timeline

Dated measurement sets are stored for each patient, and the twin state is recomputed whenever new measurements are entered. The observed history (re-assessed states) is displayed separately from the

simulated projections of Layer 5, which keeps the distinction in §1.1 explicit while implementing the "updating with new information" property claimed in §14.

17.6 Recommendation diversity and weekly meal plans

Diversity. Near-duplicate recipes (for example, several chickpea dishes) are reduced with maximal-marginal-relevance re-ranking on ingredient sets.
Meal plans. A seven-day plan of two main meals a day is built so that every meal and every day meets the evidence-linked reference values, protein sources vary within each day and across the week, and practical recipes come first. Because a patient may not have the ingredients, tools or taste for another cuisine, the week is filled in steps: the patient's own cuisine first; then a favourite recipe may repeat, never on consecutive days; and only then recipes from the most similar cuisines by ingredient profile, each labelled as borrowed.

17.7 Grounded plain-language summaries

A language model writes a short patient-friendly summary using only the structured twin output (state, priorities, evidence IDs and recipe values). An automatic check confirms that every number and citation in the text appears in that output; if it fails, a template summary is shown instead. This follows recent guidance that clinical language-model output be grounded and validated.

17.8 Fairness audit

Layer 2 performance and Layer 4 criteria satisfaction are reported by sex, race/ethnicity and age group with 95% bootstrap intervals.

17.9 External and long-term validation with NHANES III (new data)

Data. Archived NHANES III (1988–1994) gallbladder ultrasound images were re-read in 2009–2010 to grade hepatic steatosis in adults aged 20–74, and NHANES III participants are linked to the National Death Index through 31 December 2019 in the public-use Linked Mortality Files.
Use. This gives (a) external validation of the twin against a different reference standard in a different era and (b) the project's first outcome data: whether the twin's steatosis risk and FIB-4 are associated with all-cause and liver-related mortality. It partly answers the cross-sectional limitation in §1.1 without requiring repeat imaging.
Caveats. Assays, measurement protocols and the population have changed since 1994, some inputs may be unavailable, and the public mortality files perturb follow-up time or cause of death for selected records.

17.10 Ultra-processing criterion (new data)

The EASL-EASD-EASO guideline advises limiting ultra-processed foods, but the ranking cannot yet measure processing. FoodProX (Menichetti et al., 2023) estimates the NOVA processing level from nutrient profiles and was trained on FNDDS; applied to recipe ingredients, it supports a new "limit ultra-processed foods" criterion linked to the existing guideline evidence row.

17.11 Glycaemic load criterion (new data)

The International Tables of Glycemic Index and Glycemic Load Values 2021 (Atkinson et al., 2021; more than 4,000 items) can be mapped to recipe carbohydrate ingredients to estimate glycaemic load, supporting the glucose-related priority for patients with elevated HbA1c. An evidence row linking glycaemic load to liver outcomes is required before it is weighted.

17.12 Recipe acceptability

Acceptability. RecipeDB ratings and rating counts, already parsed, can serve as an acceptability signal, so that well-rated recipes are preferred among those with comparable health scores.

17.13 Additional evidence rows

Candidate rows include coffee, which observational meta-analyses associate with lower odds of advanced fibrosis (OR 0.73; Liu et al., 2015) and significant fibrosis (Ebadi et al., 2021) but not with NAFLD incidence; fasting interventions, which reduced ALT in a 2026 meta-analysis of RCTs (Stern et al., 2026); and the null findings for omega-3 supplementation and low-carbohydrate high-fat diets from the same review. Observational associations would be labelled as such and not converted into causal effect sizes.

17.14 Future work

Food–disease and spice–disease association resources such as DietRx and SpiceRx could add an ingredient-level health-relevance signal to the ranking engine. Because these associations are largely literature-mined and observational, they would be kept separate from the graded intervention evidence and labelled as associations rather than effects.

The patient cohort is drawn from the US population, so the recipe corpus focuses on the cuisines eaten there. The 20,000 RecipeDB recipes available to the project include only 19 Indian recipes; access to the full RecipeDB, including Indian regional cuisines, would allow the twin to be validated for Indian users, for whom the Asian BMI thresholds are already implemented.

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